

Sumin Cho

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Research Interests

Animatable 3D/4D Avatars, Digital Humans, Deformable 3D Reconstruction, Neural Rendering, Controllable 3D Generative Models.

Education

Korea Advanced Institute of Science and Technology (KAIST), South Korea Sep 2024 – Present
(Expected: Aug 2026)

- Master of Science, Graduate School of Culture Technology.
- Visual Media Lab (Computer Graphics Lab, Prof. Junyong Noh).
- **GPA:** 4.1/4.3. **Relevant Coursework:** Computer Vision, Computer Graphics, Deep Learning.

University of Georgia (Exchange Student), United States Aug 2022 – Dec 2022

- Non-degree exchange program; coursework credited toward B.S. degree.

Sogang University, South Korea Mar 2019 – Aug 2024

- Bachelor of Science in Art & Technology, Software, Media & Entertainment (Triple Major).
- **GPA:** 3.55/4.3.

Research Experience

A Training-Free Framework for View-Coherent 3D Stylization (Under Review, SIGGRAPH Asia 2026), KAIST Jul 2025 – Present

- Proposed a representation-agnostic, training-free 3D stylization framework that ensures multi-view consistency by injecting self-attention outputs from a stylized anchor view into arbitrary target views.
- Developed a plug-and-play pipeline compatible with various 3D representations (NeRF, 3DGS, and Mesh), enabling consistent style transfer for unseen viewpoints, such as top-down views, using only a single reference style image.

Generative Pipeline for Garment Mesh Generation (Under Review, SIGGRAPH Asia 2026), KAIST Nov 2024 – Present

- Developed a text-guided garment generation pipeline for stylized full-body characters, using a DGCNN-based 3D mask estimator to guide garment deformation and generate character-fitted garment meshes.

Sensory Kitchen: AI-Based Cooking Education System for Visually Impaired, KAIST Sep 2024 – Dec 2024

- Developed multimodal cooking assistant using generative AI and tactile interaction.

ConTEXTure: Consistent Multi-view Texture Generation for 3D Meshes (arXiv preprint), Sogang Univ. Feb 2024 – Jul 2024

- Developed simultaneous multi-view texture generation pipeline addressing viewpoint bias.

Projects

IllusionCraft: Diffusion-Optimized Textures for Transforming Surfaces, KAIST Oct 2024 – Dec 2024

- Built multi-view optimization pipeline embedding multiple distinct images into a single 3D object surface to create view-dependent 3D visual illusions.

- Few-Step Diffusion Model for Simpsons Dataset**, KAIST Apr 2024 – Jun 2024
- Trained an EDM-style teacher and Consistency Distillation student for NFE=1/2/4 fast sampling, achieving competitive FID with drastically fewer evaluations.
- 3D Virtual Indoor Scene Generation via LLM and RAG**, KAIST; KAI Oct 2024 – Nov 2025
- Developed automated 3D scene generation system (registered Korean software copyright).
 - Implemented core technology for an industry demo exhibited at SIGGRAPH 2025.
- Stylized Infinite-Zoom 3D Scene Generation**, KAIST Oct 2024 – Dec 2024
- Built full 3D infinite-zoom pipeline combining Gaussian Splatting reconstruction, 3D style transfer, and seamless scene blending for continuous zoom transitions.

Industry Experience

- Software Engineer (Freelance)**, Abear (Windly) Jul 2024 – Feb 2026
- Building an automated product image processing pipeline for e-commerce platforms using web crawling and computer vision-based image editing tools.
- Media AI Lab Researcher**, Sogang University Mar 2023 – Jul 2024
- Conducted research on computer vision and 3D content generation using diffusion models.
- Founder & Product Developer**, Handy (Foot-tech Startup) Mar 2021 – Dec 2022
- Founded a tech startup; designed a 3D foot-measurement device and implemented an end-to-end web workflow for personalized footwear recommendations.

Honors and Awards

- Certificate of Excellence** – Outstanding TA for Education 4.0Q Program, KAIST Sep 2025
- Excellence Award** – Samsung Securities Co. Digital and IT Academic Program Dec 2023

Professional Service & Leadership

- **Reviewer**, ACM SIGCHI Conference on Human Factors in Computing Systems 2026 Posters.
- **Graduate Student Council President**, KAIST — Led academic initiatives and coordinated communication between graduate students and faculty.
- **Exhibition Staff**, CES 2024 & SIGGRAPH Asia 2025 & SIGGRAPH 2026.

Additional Training

- **Naver Boostcamp AI Tech** – Computer Vision Track (2023-2024).
- **Google AI Bootcamp** – Computer Vision Specialization (2023).

Technical Skills

Programming: Python, C++, CUDA | **ML/Graphics:** PyTorch, Diffusers, Nerfstudio
3D Tools: Unity, Maya | **DevOps:** Linux, Git